

Ballot words and the Morin Thom polynomials

Chern++

September 2026

Abstract

The Chern coefficients $C(M)$ of the Morin Thom polynomials Tp_{A_d} are indexed by zero-sum multisets M of d integers. Let $b(M)$ be the number of orderings of M whose proper partial sums are nonnegative. We prove, given two geometric inputs that we state precisely, that $C(M) = b(M)$ whenever $\max M \leq 1$. On that sector $b(M)$ is a Kreweras number. We conjecture that $C(M) \geq b(M)$ for every M , with equality exactly on that sector. This implies Rimányi's positivity conjecture, and it holds, with its equality case, on every exact coefficient we can compute: all $d \leq 7$ on large boxes, including Tp_{A_7} to $\ell = 5$ and every packet of charge ≤ 6 at $d = 7$.

1 Setting

By Rimányi's ℓ -free reduction [2] and the Bérczi–Szenes residue formula [1],

$$\text{Tp}_{A_d}^\ell = \sum_M C(M) \prod_{a \in M} c_{\ell+1+a},$$

where M ranges over multisets of d integers summing to 0, $c_0 = 1$, and $c_j = 0$ for $j < 0$. The coefficient $C(M)$ does not depend on ℓ . In chamber coordinates,

$$C(M) = \sum_{\alpha} A_{\beta(\alpha)},$$

where the sum is over the distinct orderings α of M , the prefix sums $\beta_j = \alpha_1 + \cdots + \alpha_j$ ($j < d$) must be nonnegative, and $F_d = \sum_{\beta} A_{\beta} x^{\beta}$ is the Morin chamber series. Rimányi's conjecture is $C(M) \geq 0$.

Definition 1. A *ballot ordering* of M is an ordering whose proper prefix sums are all nonnegative. Let $b(M)$ be the number of distinct ballot orderings. Equivalently, $b(M)$ is $C(M)$ computed with the pure chain series $\prod_{j=1}^{d-1} (1 - x_j)^{-1}$, all of whose coefficients are 1, in place of F_d .

The decreasing ordering is always ballot, so $b(M) \geq 1$. The *charge* of M is $\max M$. The *plane sector* is charge ≤ 1 ; for $\ell = L - 1$ these are the Chern monomials whose largest index is at most $\ell + 2$.

2 The plane sector

Theorem 2. Assume the geometric inputs (I1) and (I3) of §2.1. If $\max M \leq 1$ then $C(M) = b(M)$. Write $M = (1^{d-b}, 1 - s_1, \dots, 1 - s_b)$ with every $s_i \geq 1$, and let m_i be the number of s 's equal to i . Then this common value is the number of noncrossing partitions of $[d]$ with block sizes $s_1, \dots, s_b$:

$$C(M) = b(M) = \frac{d!}{(d - b + 1)! \prod_i m_i!}.$$

Independently of (I1) and (I3), the identity $C(M) = b(M)$ is verified by exact computation from the chamber series on every plane-sector packet with $d \leq 7$.

2.1 The geometric inputs

Fix $n = d + 1$ and $r \geq 2$.

- Let $C_{n,r}$ be the *curvilinear component* of the punctual Hilbert scheme $\text{Hilb}_0^n(\mathbb{C}^r)$. This is the closure of the locus of length- n subschemes supported at the origin and isomorphic to $\text{Spec } \mathbb{C}[t]/(t^n)$. It is irreducible of dimension $(r - 1)d$.
- Let V be the tautological rank- n bundle, whose fibre at Z is $\mathcal{O}_Z$.
- Let $E = \ker(V \rightarrow \mathcal{O})$ be its rank- d augmentation part, with Chern roots $\epsilon_1, \dots, \epsilon_d$.

Write the Thom polynomial in the Chern classes $c_i = e_i(x_1, \dots, x_q)$ of formal roots $x_1, \dots, x_q$, where $q = r + \ell$; taking r large enough makes q exceed every index that occurs.

$$(I1) \text{ Incidence formula. } \text{Tp}_{A_d}^\ell = \int_{C_{n,r}} \prod_{a=1}^q \prod_{i=1}^d (x_a + \epsilon_i).$$

This is the curvilinear form of the Bérczi–Szenes theorem [1]. Reason: a q -tuple of function germs on $\mathbb{C}^r$ defines an A_d singularity exactly when its zero scheme is curvilinear of length n . The incidence variety of pairs (curvilinear Z , q -tuple vanishing on Z) is a vector bundle over $C_{n,r}$, namely the kernel of evaluation onto $E \otimes \mathbb{C}^q$. Its projection to the space of q -tuples is generically one-to-one onto the A_d locus, which has codimension $qd - (r - 1)d = d(\ell + 1)$. The pushforward is the displayed integral. In exactly this form, with this proof, it is stated in [3].

- (I2) *Extraction.* Let $\lambda = (\lambda_1, \dots, \lambda_s)$ have all parts $\leq q$ and $s \leq d$, and let $\nu = \text{sort}(q^{d-s}, q - \lambda_1, \dots, q - \lambda_s)$. Then the coefficient of the Chern monomial c_λ in (I1) is $\int_{C_{n,r}} m_\nu(\epsilon)$, where m_ν is the monomial symmetric function.

Proof. $\prod_i \prod_a (x_a + \epsilon_i) = \prod_i \sum_{j=0}^q e_j(x) \epsilon_i^{q-j}$. The coefficient of $e_{\lambda_1}(x) \cdots e_{\lambda_s}(x) e_0(x)^{d-s}$ is the sum over distinct assignments of the indices $\lambda_1, \dots, \lambda_s, 0^{d-s}$ to the d roots, that is, $m_\nu(\epsilon)$. $\square$

- (I3) *Hyperplane lemma.* $c_d(E) \cap [C_{n,r}] = [C_{n,r-1}]$, where $C_{n,r-1}$ is embedded as the subschemes lying in a coordinate hyperplane.

Reason: the last coordinate u_r gives a section of E . It vanishes exactly on the subschemes contained in $\{u_r = 0\}$. On the standard curvilinear chart $I = (u_2 - p_2(t), \dots, u_r - p_r(t), t^n)$ it is the regular d -tuple of coefficients of p_r , so the zero locus is reduced of the expected codimension d . This is stated and proved in this form in [3].

- (I4) *The plane* (classical). For $r = 2$, $C_{n,2} = \text{Hilb}_0^n(\mathbb{C}^2)$ by Briançon's theorem [4]. Its cohomology is identified by Lehn–Sorger [5] with the associated graded of the class algebra of S_n (filtered by n minus the number of cycles). Under this identification the elementary symmetric functions of the ϵ_i go to those of the Jucys–Murphy elements $J_k = \sum_{a < k} (a k)$. The top degree is spanned by the class of the n -cycles, normalised by $\int c_d(E) = 1$. So for $|\nu| = d$, $\int_{C_{n,2}} m_\nu(\epsilon) = [\sigma] m_\nu(J_1, \dots, J_n)$ for any n -cycle σ .

Reduction to the plane. Take $\ell = L - 1$ and a plane-sector M , so the Chern monomial is c_λ with $\lambda_i = L + \alpha_i \leq L + 1$ for the entries α_i of M with $L + \alpha_i > 0$. With $q = r + L - 1$, every entry of ν is at least $q - L - 1 = r - 2$. While $r > 2$ all entries are positive, so $m_\nu(\epsilon) = (\epsilon_1 \cdots \epsilon_d) m_{\nu-1}(\epsilon) = c_d(E) m_{\nu-1}(\epsilon)$. By (I3) this lowers r by one and every entry of ν by one. At $r = 2$, $q = L + 1$ and the entries of ν are $L + 1 - \lambda_i = 1 - \alpha_i$, together with $q = 1 - (-L)$ for the entries $\alpha = -L$ that give c_0 . So ν is the multiset $1 - M$, and by (I4)

$$C(M) = [\sigma] m_{1-M}(J_1, \dots, J_n).$$

Everything below is new.

Lemma 3. Let $\sigma(x) = x + 1 \pmod{n}$. A minimal monotone factorisation of σ is an expression $\sigma = (a_1 b_1)(a_2 b_2) \cdots (a_{n-1} b_{n-1})$ with $a_i < b_i$ and $b_1 \leq \cdots \leq b_{n-1}$. Its exponent vector is $e_k = \#\{i : b_i = k\}$, with $e_1 = 0$. Then for every $e \in \mathbb{Z}_{\geq 0}^n$ with $\sum e_k = n - 1$:

$$\#\{\text{minimal monotone factorisations of } \sigma \text{ with exponent vector } e\} = \begin{cases} 1 & \text{if } D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0 \text{ for all } j \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Necessity. The $n - 1$ transpositions of a minimal factorisation of an n -cycle are the edges of a tree on $[n]$. Those with $b_i \leq j$ are edges of a forest on $[j]$, so there are at most $j - 1$ of them.

Existence and uniqueness, by induction on n . The case $n = 1$ is trivial. Let $m = e_n$ and let $(a_1 n) \cdots (a_m n)$ be the final m factors, with product C . The remaining factors form a minimal monotone factorisation of $F = \sigma C^{-1}$, a permutation of $[n - 1]$ with exactly m cycles. Since $C^{-1} = (n a_1 a_2 \cdots a_m)$ we have $a_1 = \sigma^{-1}(n) = n - 1$ and

$$F(x) = x + 1 \ (x \notin \{a_i\}), \quad F(a_i) = a_{i+1} + 1 \ (i < m), \quad F(a_m) = 1.$$

Because $F \cdot C = \sigma$ is one cycle and C is an $(m + 1)$ -cycle, each cycle of F contains exactly one a_i . Starting at $a_{i+1} + 1$ and stepping by $+1$, the cycle through a_i must meet a_i before any other a_j . So $a_{i+1} < a_i$ with no a_j in between, and a_m is the least. Hence $a_1 > a_2 > \cdots > a_m$, and F is the product of the increasing cycles on the consecutive intervals

$$I_m = [1, a_m], \quad I_{m-1} = [a_m + 1, a_{m-1}], \quad \dots, \quad I_1 = [a_2 + 1, n - 1].$$

Transpositions on disjoint intervals commute, so monotone factorisations of F are exactly shuffles, merged by b , of monotone factorisations of the interval cycles. Minimality on each interval says the exponents inside I_{m-i} sum to $|I_{m-i}| - 1$. Since D changes by $1 - e_k \leq 1$ per step, this means $D(a_{m-i}) = i$, and the ballot condition inside the interval means $D > i$ strictly after a_{m-i} , up to the next cut. So a_{m-i} is the last j with $D(j) = i$. When e is ballot these last visits exist, are increasing in i , and end at $D(n - 1) = m - 1$. So the cut points exist and are forced, and the induction applies to each interval (each is again an increasing cycle, of smaller length). $\square$

Proof of Theorem 2. Expanding $m_\nu(J_1, \dots, J_n) = \sum_e \prod_k J_k^{e_k}$ over distinct rearrangements e of ν (padded by zeros, $e_1 = 0$ since $J_1 = 0$), the coefficient of σ counts pairs $(e, \text{minimal monotone factorisation with exponent vector } e)$. By Lemma 3 this is the number of ballot vectors e . Setting $\alpha_k = 1 - e_{k+1}$ ($k = 1, \dots, d$) turns the condition $D \geq 0$ into $\alpha_1 + \cdots + \alpha_j \geq 0$, and e into an ordering of M . So $C(M) = b(M)$. For the product formula, apply the cycle lemma to $-M$ with a final -1 appended: n integers ≥ -1 summing to -1 , exactly one cyclic rotation of which has all proper prefix sums nonnegative. Reversal identifies these rotations with ballot orderings of M , so $b(M) = \frac{1}{n} \cdot \frac{n!}{(d-b+1)! \prod_i m_i!}$. This is Kreweras' count [6]. $\square$

Lemma 3 is checked exhaustively for $n \leq 7$; the totals are the Catalan numbers 1, 2, 5, 14, 42, 132. The identity $C(M) = b(M)$ is checked on every plane-sector packet for $d \leq 7$ directly from the chamber series. Both checks are in tier 12 of the test suite.

Remark 4. On the plane sector the Morin chamber series is therefore indistinguishable, packet by packet, from the pure chain series $\prod_j (1 - x_j)^{-1}$. For example, at $d = 7$ the packet $M = (1, 1, 1, 0, -1, -1, -1)$ has 35 ballot orderings. Eight of them carry nonzero A_β , namely 1, 4, -1, -4, 5, 6, 6, 18, and these sum to 35.

3 The ballot conjecture

Conjecture 5 (ballot). *For every zero-sum multiset M of d integers, $C(M) \geq b(M)$, with equality if and only if $\max M \leq 1$. Equivalently: the series $F_d - \prod_j (1 - x_j)^{-1}$ has nonnegative packet sums, and they vanish exactly on the plane sector.*

Corollary 6. *Conjecture 5 implies Rimányi’s conjecture, and more: every Chern coefficient of Tp_{A_d} is at least 1 whenever the monomial occurs.*

This explains a feature of the data that had no explanation: across $d = 4, \dots, 7$ the least Chern coefficient is 1, never 0. It is attained exactly at $M = 0$ and $M = (1^{d-1}, 1 - d)$, the two multisets with $b(M) = 1$ on the plane sector.

Table 1: The ballot conjecture on every exact table. A violation is $C(M) < b(M)$; the equality columns test that $C = b$ holds exactly on the plane sector.

d	box	packets	violations	plane $C \neq b$	off-plane $C = b$	$\min_{\max M \geq 2} C/b$
2	level box L=14	15	0	0	0	2
3	level box L=14	169	0	0	0	2
4	level box L=12	1033	0	0	0	9/4
5	level box L=12	7166	0	0	0	12/5
6	level box L=11	29941	0	0	0	79/30
7	level box L=6	8033	0	0	0	58/21
7	charge box p=6	8033	0	0	0	58/21

Evidence. Table 1 tests the conjecture on every exact table available. The data come from the exact sweep of `chernpp.bboxes`, reconstructed by CRT and verified against a held-back prime. The level box $\min M \geq -L$ contains every packet of Tp^ℓ with $\ell < L$; the charge box contains every packet with $\max M \leq p$. There is no violation, no plane-sector packet with $C \neq b$, and no off-plane packet with $C = b$. Off the plane sector $C(M)/b(M) \geq 2$ throughout. The minimum is 2 at $(2, -1, -1)$ and grows slowly with d (9/4, 12/5, 79/30, 58/21), attained on the family $(2, 1^{d-4}, -1^{d-2})$ and its neighbours.

What does not work.

- *Factorial weights.* $C(p, -1^p) = p!$, which suggests weighting each ballot word by $\prod_{\alpha_i > 0} \alpha_i!$. That bound fails already at $d = 3$.
- *Fusion.* Replacing a 2 by two 1s and counting plane-sector ballot words of the longer multiset gives some families exactly. For example, $C(2, 0^{d-2}, -2) = \binom{d+1}{3} + \binom{d}{2}$, where both terms are plane-sector counts. But the residual is negative elsewhere.
- *Local step weights.* Keep weight 1 on steps ≤ 1 and give each step of size 2 a weight depending on local data: its height, together with the neighbouring steps, the number of remaining steps, or the next return. Fitted exactly on the packets with one 2, every such model that is consistent within a single d is inconsistent across d .
- *Formal Jucys–Murphy extension.* Theorem 2 reads $C(M) = \sum_{\alpha} [\sigma] \prod_k J_{k+1}^{1-\alpha_k}$ over ballot α . At charge 2 this needs J_k^{-1} , which exists for $k = 2, 3$ (their eigenvalues, the contents of boxes 2 and 3, are nonzero). Where it can be evaluated, it returns the wrong values, some of them fractional ($C(2, 0, -2) = 7$ against $23/2$). At $d = 4$ the needed inverses do not exist.
- *Individual coefficients.* No bound on single A_β can work: they are unbounded below (-3622 at $d = 7$), and the smallest symmetrisation that clears all negatives grows with d [7].

4 Zero insertion

Appending a 0 to M adds a singleton block and multiplies the Chern monomial by $c_{\ell+1}$. For a zero-free base M_0 write $C_z = C(M_0 \sqcup 0^z)$ and $b_z = b(M_0 \sqcup 0^z)$. On the plane sector b_z is a Kreweras number, a polynomial in z . For example $b(1, -1, 0^z) = \binom{z+2}{2} = 1 + 2z + \binom{z}{2}$.

Conjecture 7 (zero insertion). *For every zero-free M_0 , the sequences C_z and $C_z - b_z$ have nonnegative forward differences of every order at $z = 0$:*

$$C_z - b_z = \sum_{k \geq 0} N_k(M_0) \binom{z}{k}, \quad N_k(M_0) \geq 0.$$

This refines Conjecture 5, since every M is some $M_0 \sqcup 0^z$. It holds on all 4925 zero-free bases available (`results/zero_insertion.json`), for C and for $C - b$, at every order that fits in $d \leq 7$. Where the differences terminate within range, C_z is a polynomial:

$$C(2, -2, 0^z) = 2 + 5z + 4 \binom{z}{2} + \binom{z}{3}, \quad C(3, -3, 0^z) = 4 + 13z + 16 \binom{z}{2} + 9 \binom{z}{3} + 2 \binom{z}{4}.$$

The degree equals $\sum_{a>0} a + \#\{a < 0\}$ in all four certified cases. That is too few to conjecture.

5 Where a proof has to go

In the geometric model of §2.1, a packet of charge p reduces by (I3) only to the curvilinear locus $C_{n,p+1}$ in $\mathbb{C}^{p+1}$. The plane case $p = 1$ is Theorem 2. Conjecture 5 therefore says:

the integral of $m_\nu(E)$ over the curvilinear locus in $\mathbb{C}^{p+1}$ is at least the count of ballot words, which is what the same integral gives in the plane, and exceeds it as soon as $p \geq 2$.

Two routes suggest themselves.

1. *A degeneration.* Degenerate $C_{n,p+1}$ to a cycle containing a planar component with multiplicity one, plus effective components on which $m_\nu(\epsilon)$ integrates nonnegatively. Positivity does fail on some individual degeneration components [3], so the decomposition would have to be chosen with care.
2. *A charge-2 Jucys–Murphy calculus.* Find the analogue, for the curvilinear component of the punctual Hilbert scheme of $\mathbb{C}^3$, of the Lehn–Sorger description. In such a calculus the excess $C(M) - b(M)$ should count a positive family of *non-monotone* or *branched* factorisations.

The exact tables (`results/morin_a{6,7}.json`, 20,725 coefficients) are available to test either route.

References

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